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(54) **COMPRESSED NATURAL GAS VEHICLE
SAFETY SYSTEM AND METHOD**

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(63) Continuation of application No. 17/316,170, filed on
May 10, 2021, now Pat. No. 11,524,639, which is a
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(57) **ABSTRACT**

(51) **Int. Cl.**
B60R 16/023 (2006.01)
B60R 16/08 (2006.01)
(Continued)

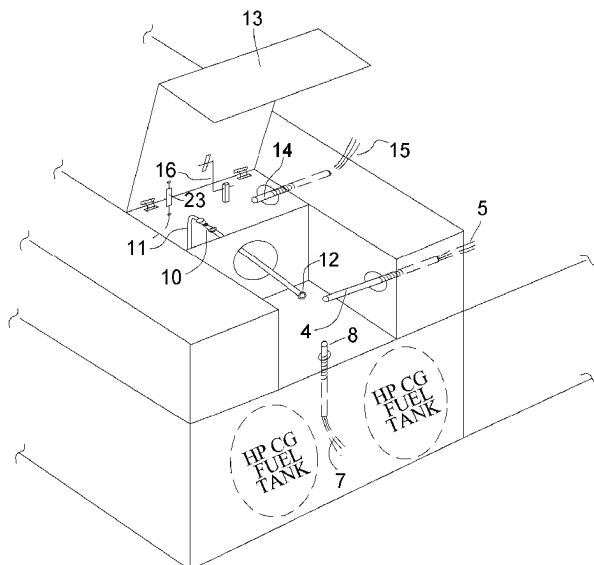
A multiply-redundant system that prevents a driver from
starting and/or moving a vehicle if a compressed natural gas
fill system is not correctly and completely disconnected
from the vehicle. One or more sensors in combination with
one or more optional microswitches combine to lock-out the
vehicle's ignition or otherwise prevent it from starting
and/or moving. For different levels of safety, different com-
binations of sensors can be used with the lowest level having
a single proximity sensor sensing the presence or absence of
a high-pressure fill hose. The highest level of safety being
achieved by having separate proximity sensors on the fuel
fill hose fitting, the gas cap cover and a manual safety valve
along with a redundant microswitch. An optional override
that may be restricted as to the number of times it can be
used can allow starting with a faulty sensor in order to allow
maintenance.

(52) **U.S. Cl.**
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CPC ... B60K 28/10; B60K 2028/003; F17C 13/02;
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See application file for complete search history.

15 Claims, 6 Drawing Sheets



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F17C 5/06 (2006.01)

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CPC **B67D 7/3281** (2013.01); **F17C 5/06** (2013.01); **F17C 13/02** (2013.01); **B60Y 2300/436** (2013.01); **B60Y 2300/70** (2013.01); **B60Y 2302/01** (2013.01); **B60Y 2306/13** (2013.01); **B60Y 2400/301** (2013.01); **B60Y 2400/3019** (2013.01); **F17C 5/007** (2013.01); **F17C 2205/0329** (2013.01); **F17C 2205/0332** (2013.01); **F17C 2205/037** (2013.01); **F17C 2221/033** (2013.01); **F17C 2223/0123** (2013.01); **F17C 2223/0153** (2013.01); **F17C 2223/036** (2013.01); **F17C 2227/0157** (2013.01); **F17C 2250/032** (2013.01); **F17C 2250/034** (2013.01); **F17C 2250/036** (2013.01); **F17C 2250/0452** (2013.01); **F17C 2250/0465** (2013.01); **F17C 2250/0478**

(2013.01); **F17C 2260/042** (2013.01); **F17C 2265/065** (2013.01); **F17C 2270/0168** (2013.01); **Y10T 137/8158** (2015.04)

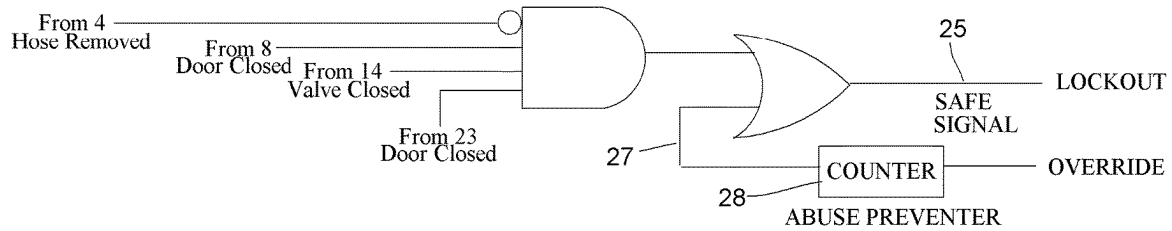
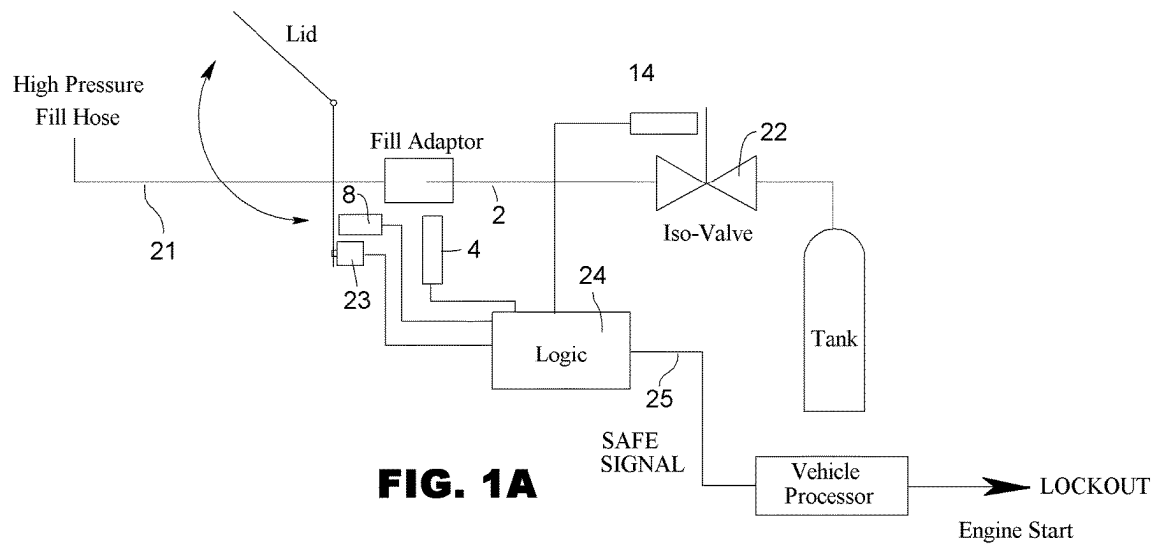
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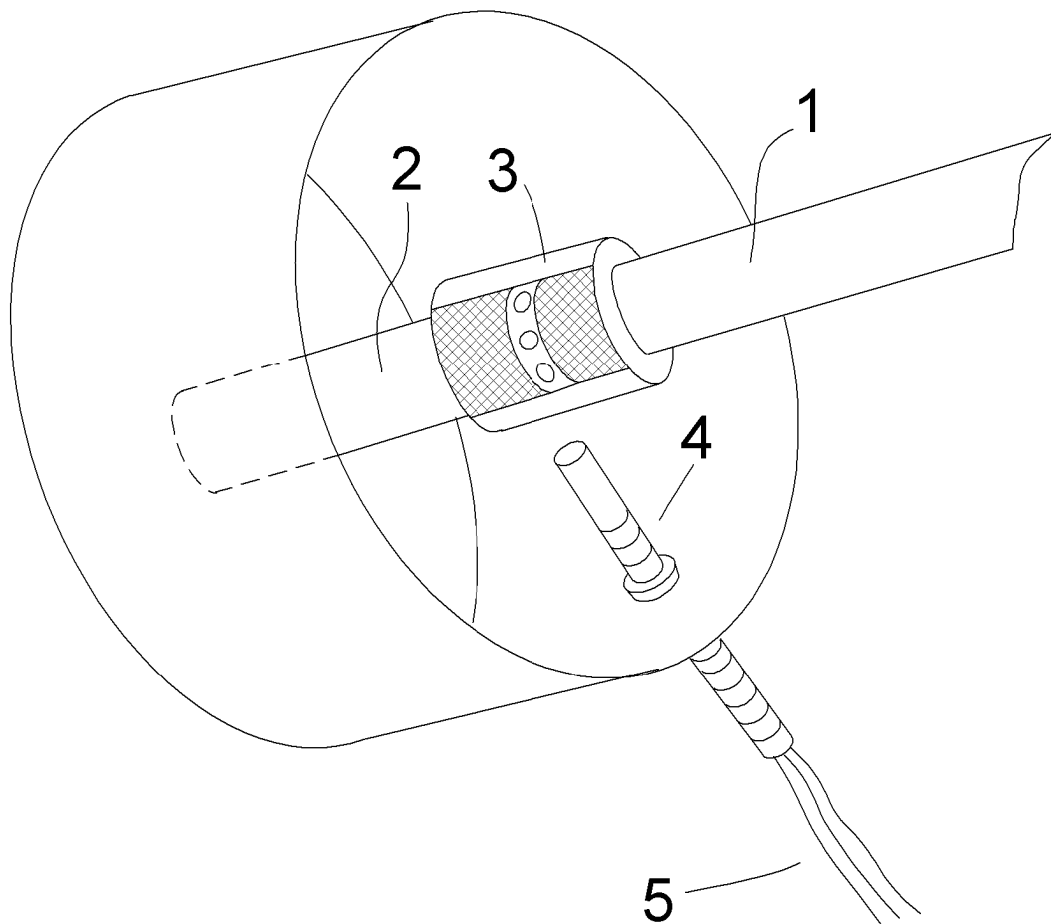
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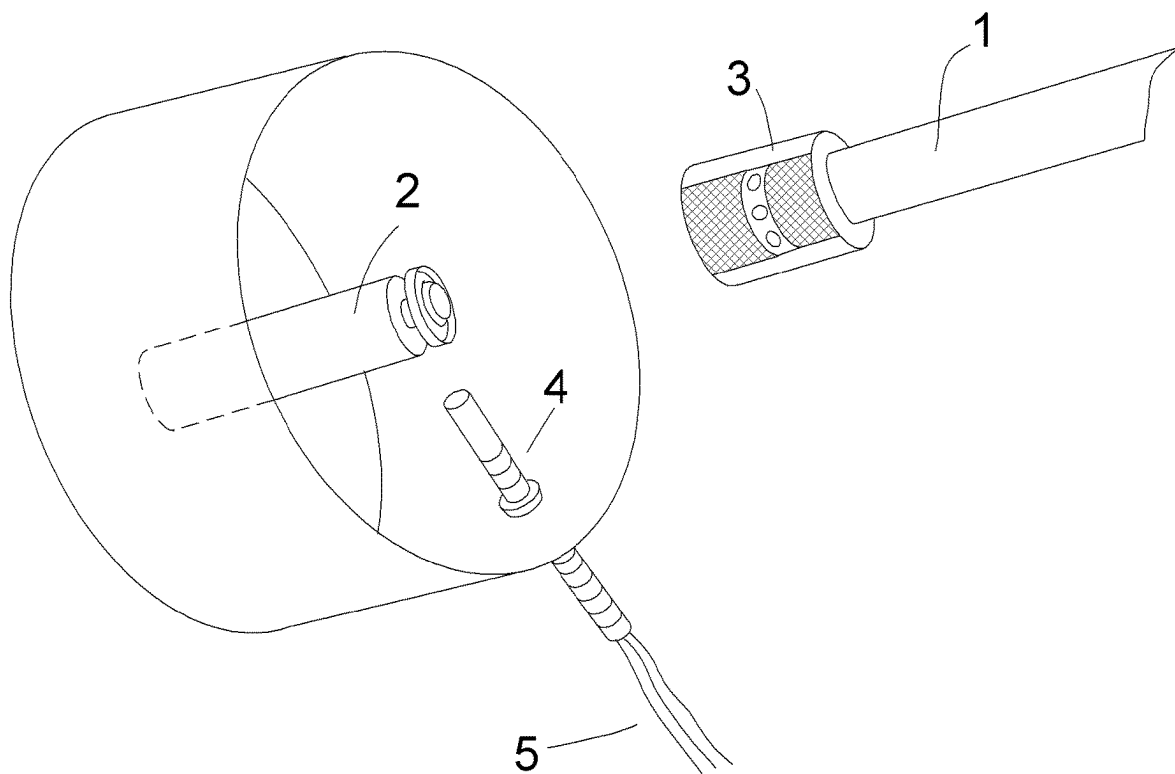
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**FIG. 2**

**FIG. 3**

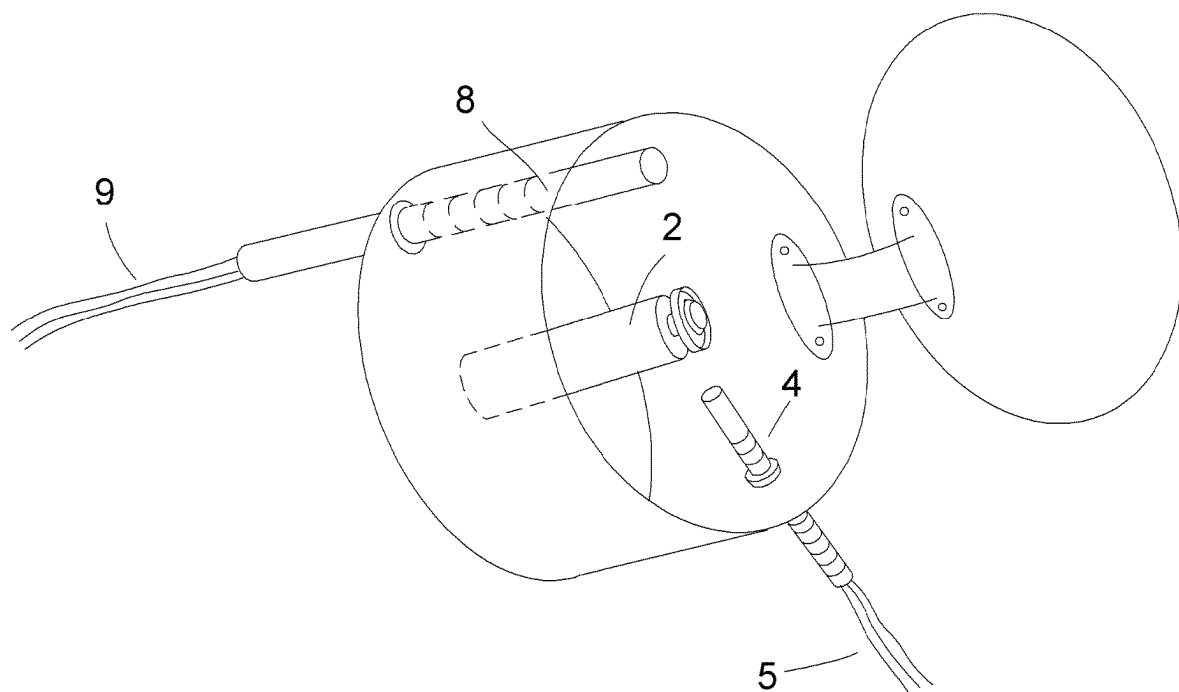
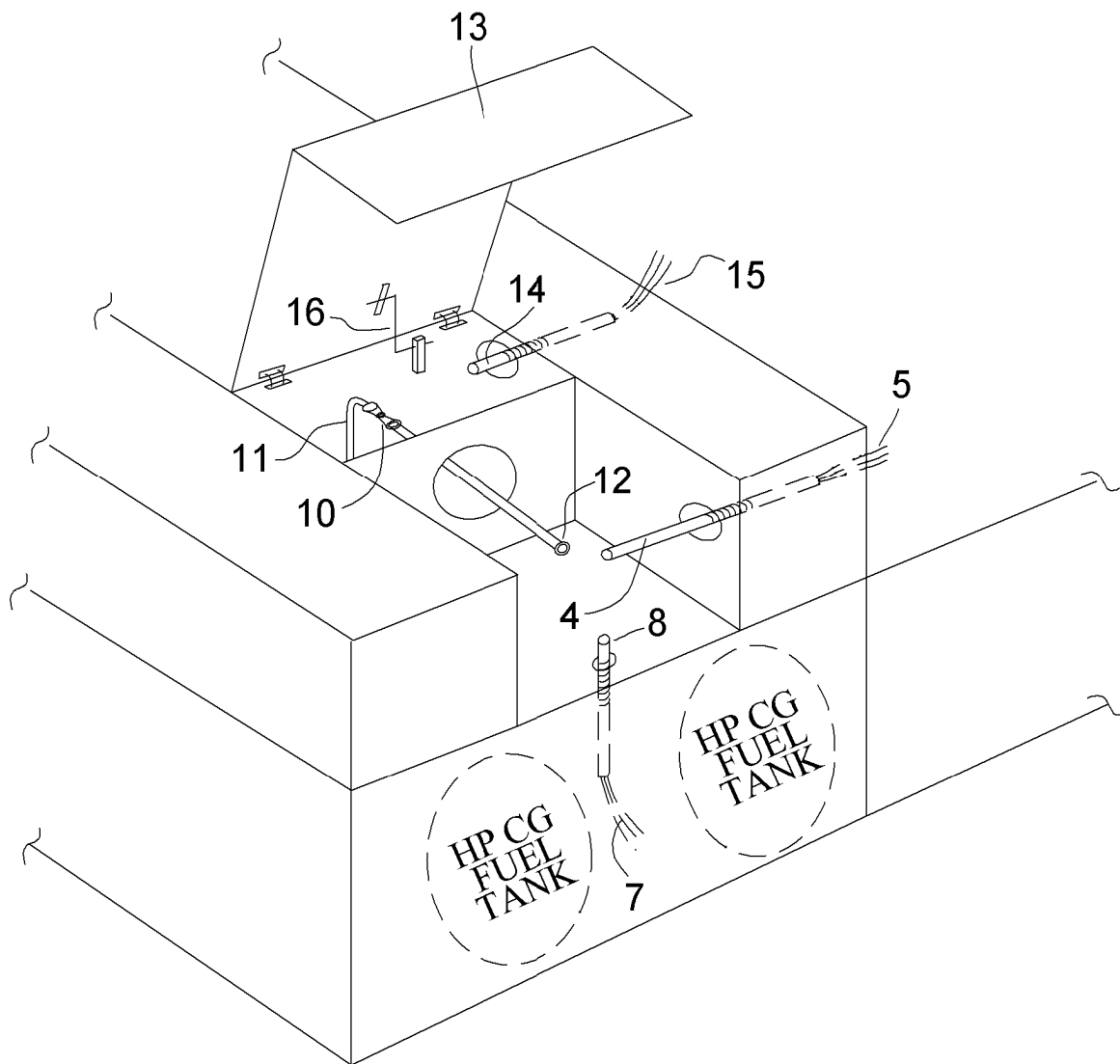


FIG. 4

**FIG. 5**

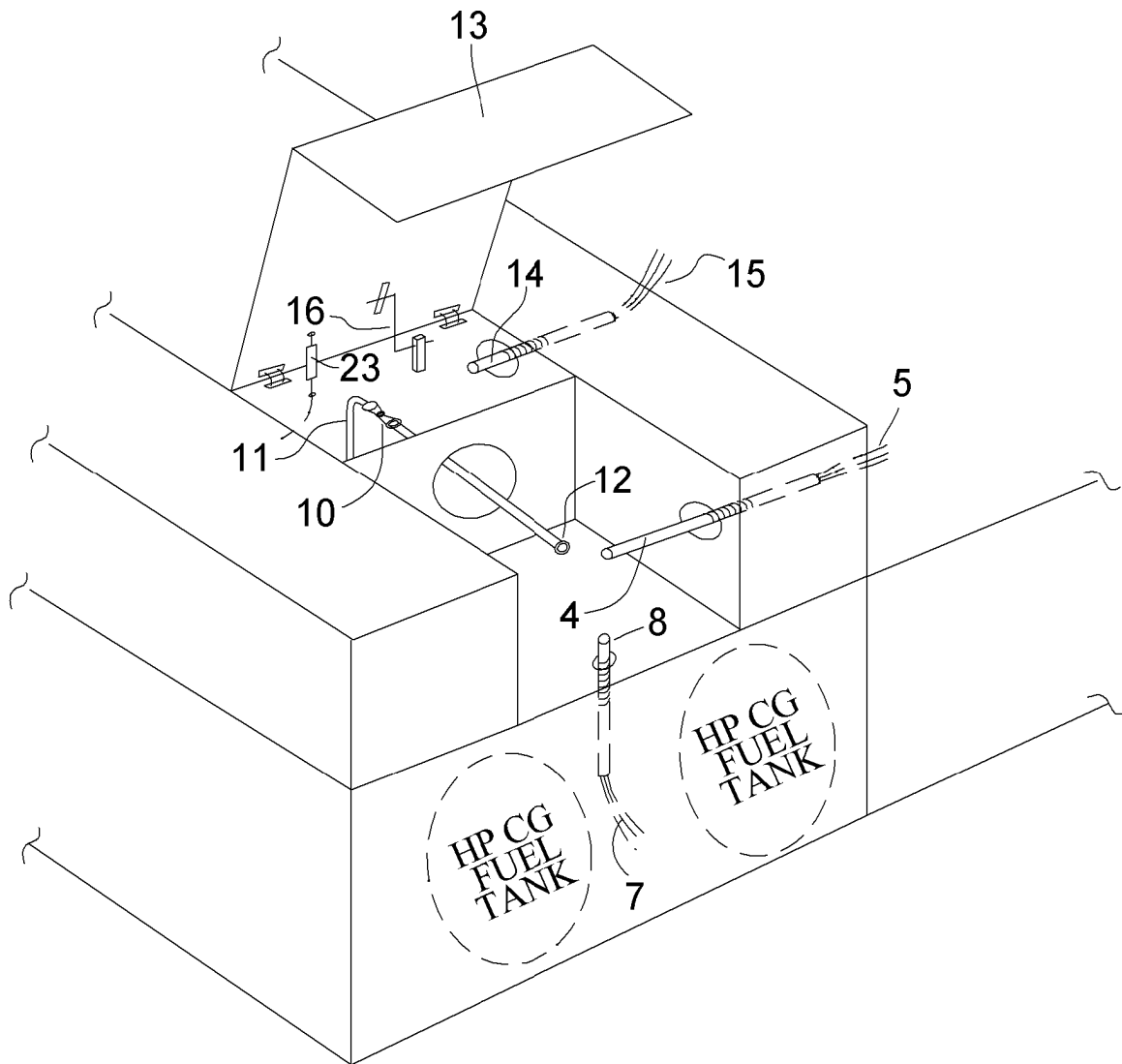


FIG. 6

COMPRESSED NATURAL GAS VEHICLE SAFETY SYSTEM AND METHOD

This is a continuation of Ser. No. 17/316,170 filed May 10, 2021, now U.S. Pat. No. 11,524,639 issued Dec. 13, 2022 which was a continuation of application Ser. No. 15/925,891 filed Mar. 20, 2018 which was a continuation of application Ser. No. 15/257,030 filed Sep. 6, 2016, now U.S. Pat. No. 9,919,663 issued Mar. 20, 2018 which was a continuation of application Ser. No. 14/194,131 filed Feb. 28, 2014, now U.S. Pat. No. 9,434,329 issued Sep. 6, 2016 which was a continuation of application Ser. No. 13/463,022 filed May 3, 2012, now U.S. Pat. No. 8,662,235 issued Mar. 4, 2014, application Ser. No. 17/316,170, Ser. No. 15/925,891, Ser. No. 15/257,030, Ser. No. 14/194,131 and Ser. No. 13/463,022 are hereby incorporated by reference in their entireties.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates generally to the fields of vehicle safety and compressed natural gas and more particularly to safety system related to fueling vehicles powered by compressed natural gas.

Description of the Prior Art

Natural gas is becoming an abundant resource in the United States and several other countries. It is estimated that the natural gas reserves of the U.S. exceed the oil reserves of Saudi Arabia in terms of years of energy supply.

In order to utilize this resource effectively, and to replace crude oil, natural gas must be used in vehicles. Many fleet operators are switching their vehicles to natural gas since, with minor modifications, internal combustion engines run very well on natural gas.

Natural gas is typically supplied in two different forms: 1) as a compressed gas, and 2) as a liquefied gas. While more efficient in terms of the amount of gas that can be supplied in a single tank, liquefied gas is very dangerous to handle and requires highly specialized equipment to fuel with it and actually use it. On the other hand, compressed natural gas is relatively easy to fuel with and to utilize. Compressed natural gas can be supplied in pressure bottles at pressures between 3000 psi and 4000 psi. It is very well known in industry how to handle and fill such bottles. A natural gas "gas station" can fill a vehicle tank (pressure bottle) in just a few moments using fill techniques similar to high pressure air. A vehicle tank containing 100-200 cu. feet of natural gas at a pressure of 3500 psi is typically not more dangerous than a typical automobile's gasoline tank. The tank can be shielded from direct impact in an accident.

Many households and businesses in the U.S. have piped in natural gas for cooking and, in many cases, heating. Most Americans own at least one car or other vehicle, with many households owning several vehicles. This combination immediately suggests the possibility of filling a natural gas vehicle at home during the night for use the next day, or filling at the location of a small business. Because even a large compressed natural gas tank will not last as long between refills as a typical gasoline tank, it will become a very convenient to refill it during the night. The vehicle owner can also use natural gas filling stations; however, these will probably be more convenient for extended trips. Most people would prefer not to have to wait 5-10 minutes

at a gas station to fill their vehicle. While exchangeable tanks are also possible, these require considerably more logistics and handling than a simple fixed onboard tank. Even with exchangeable tanks, the vehicle owner would still probably have to wait considerably longer to get a refill at a station than they currently do with gasoline.

At the present time, at least one car manufacturer is supplying a compressed natural gas vehicle and a home compressor to fill it. National fire codes currently prevent large gas storage tanks in homes or compressors that fill at fast rates. Given present standards, the home compressor will be directly attached to the vehicle via a high pressure hose, and the vehicle will take several hours to fill (depending on the size of the onboard tank). Again, this will be very conveniently accomplished during the night. The same arrangement can be used at small businesses, especially those with fleets of vehicles.

However, with numerous individuals filling natural gas vehicles in their garages at night or at their business locations, the potential for a very serious type of accident increases dramatically—namely what is called a drive-off accident. This is where the driver tries to drive away with the high pressure filling hose still attached to the vehicle. Such an accident in a home garage could be catastrophic if the fitting or valve on the vehicle is damaged, or if the compressor is pulled away from the home natural gas source. In either case, a considerably amount of gas could escape into the garage causing a fire or explosion danger. Also, even a "soft" drive-away accident, one where the driver stopped before breaking the hose or fitting could stress the fill hose and possibly cause small, very hard to detect, gas leaks. Such small leaks could result in the garage being filled with gas by morning.

Natural gas contains mostly methane and is thus lighter than air. Natural gas leaking in a garage will fill the garage from top to bottom. An explosive mixture for natural gas and air is between around 5% (for pure methane) and around 15-20%. Many garages contain furnaces and water heaters having open flames. As natural gas fills a garage, it can be easily ignited by a furnace or a water heater on a raised pedestal (fire codes require open flame devices in garages to be mounted on pedestals to avoid gasoline vapor that might collect along the floor from a car gasoline leak). This is an ideal situation for a garage explosion. Hence anything that has the possibility of causing the release of natural gas or of causing a gas leak becomes a large danger. A drive-away accident is such an event.

Some techniques have been reported in the art to prevent a gasoline vehicle from starting at a gasoline station if the fill nozzle is still in the gas tank inlet. Among these are U.S. Published Application No. 2002/0162601 and U.S. Pat. No. 5,720,327. Other techniques have been invented to prevent pumping gasoline or other fuel if the fill hose is not in the fill inlet. Among these are U.S. Published Application No. 2008/0290152 and U.S. Pat. No. 4,227,497.

While these prior art techniques are useful for gasoline, they do not solve the problems associated with home or business filling a vehicle with high pressure compressed natural gas. In particular, a compressed natural gas tank is filled to a very high pressure (between 3000 and 4000 psi). A leak or disconnect of a fitting or valve on such a pressure vessel can cause a tremendous pressure explosion that can act like a bomb (this can happen with any compressed gas including air). Also, a very small leak at high pressure can cause a large quantity of gas to escape. For example, an exploding tank can send metal fragments in all directions with enough force to penetrate the house and the driver's

3

compartment of the vehicle. This can happen before there is any fire. After that, the tiniest spark could ignite the now explosive and highly flammable gas cloud reducing the house or business to rubble. Even a quick-disconnect hose does not typically solve the problem since the driver may attempt to drive away very quickly still damaging the fittings, tank, hose, compressor and/or natural gas supply.

It would be very advantageous to have doubly or triply redundant system and method that prevents the driver from ever starting the vehicle when the filling hose is attached and a safety valve is not in the correct position.

SUMMARY OF THE INVENTION

The present invention relates to a multiply-redundant system and method for preventing a driver from starting or moving a compressed natural gas vehicle if the high pressure gas fill system is not correctly and completely disconnected from the vehicle.

In one embodiment of the invention, multiple electrical proximity sensors, or other sensors, in combination with one or more optional mechanical microswitches combine to lock-out the vehicle's ignition or otherwise disable the vehicle. For different levels of safety, different combinations of sensors can be used, with the lowest level having a single sensor sensing the presence or absence of a fuel supply fitting. The highest level of safety according to the invention is to have separate sensors, such as proximity sensors, on the fuel fill hose fitting, the gas cap cover and a manual safety or isolation valve along with a redundant microswitch on at least one of the components. The ignition, transmission or other function can be locked out by an electrical or mechanical lockout provided to the vehicle's computer by the manufacturer, or by a simple series electrical circuit in the ignition string. The safest system could use both techniques.

In addition, the vehicle's computer, or another simple electrical circuit, could provide a visual and/or audio indication that one or more of the sensors was indicating an attached fuel hose. The alarm could be activated the instant the driver inserts a key into the ignition. Under the present invention, it is understood that there is a possibility that one of the sensors might fail in a state that indicates the fuel hose is connected when, in reality, the system is entirely safe. In this very special case, the present invention provides a technique for the driver, on a limited-time basis, to override the sensors and start the engine or move the vehicle using a supplied override. Having this feature allows the driver to drive to a mechanic to have the problem fixed. This feature could be automatically disabled by an abuse preventer after a predetermined number of uses (such as three times) to keep a driver with a bad sensor from delaying getting it fixed. Finally, an optional break-away fitting can also be provided in addition to the other features of the invention for a final level of protection.

BRIEF DESCRIPTION OF DRAWINGS

Attention is now directed to several drawings the illustrate features of the present invention:

FIG. 1A shows a block diagram of an embodiment of a high-safety lock-out system according to the present invention.

FIG. 1B shows a logic circuit that can combine sensors inputs.

FIG. 2 shows a detail drawing of a fill fixture with a proximity sensor.

4

FIG. 3 shows the system of FIG. 2 with the high pressure fill hose removed.

FIG. 4 shows a system similar to that of FIGS. 2-3 with an additional gas cap sensor.

FIG. 5 shows a panel-type filling system with an isolation valve and three proximity sensors.

FIG. 6 shows the system of FIG. 5 with an additional mechanical microswitch.

Several drawings and illustrations have been presented to aid in understanding the present invention. The scope of the present invention is not limited to what is shown in the figures.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The present invention relates to a system and method that provides a lockout out to a vehicle's ignition, transmission or other means of moving the vehicle when a high pressure compressed natural gas fuel hose is attached to the vehicle an/or the fill cap is open. FIG. 1A shows a block diagram of an embodiment of such a system. A compressed gas fuel fitting 2 accepts a high pressure filling hose 21 that allows filling through an isolation valve 22. The isolation valve 22 can be manually opened, or it can be opened mechanically when a panel lid or "gas cap" is opened. A sensor 4, which can be an electrical proximity sensor, senses the presence of the fitting part of the high pressure filling hose 21. An optional second sensor 8 senses that the panel or gas cap is open. An optional third sensor 14 can sense the position of an isolation valve 22 (on vehicles that have such a valve). All of the sensors can be magnetic, optical or ultra-sonic proximity sensors, or any other sensors, and use any method of sensing proximity or that a particular mechanical part is or is not in a particular position. One or more optional mechanical microswitches 23 can provide a backup to one or more of the sensors.

A further logic circuit 24 or other lockout which can be a separate unit, or part of the vehicle's processor, can combine the inputs from all the sensors 4, 8, 14, and the optional microswitch 23 to produce a safe signal 25 that will allow the vehicle ignition to start the car, or otherwise allow the vehicle to move. FIG. 1B shows a schematic representation of this circuit 24. Here, each sensor's signal is amplified and conditioned and fed into an AND logic circuit 26. If any of the sensors is in an unsafe condition, the AND circuit 26 will not produce a "safe" signal. The particular circuit of FIG. 1B assumes that a proximity sensor has a logic high when there is proximity, and that the microswitch is high when the lid is closed. Any other logic levels or configurations can be used. In particular, a program in a microcontroller or other processor could also make the determination. As is well understood in the art, an OR circuit could also be effectively used instead of an AND circuit. Any circuit or program that combines sensor inputs to make a "safe" determination is within the scope of the present invention.

FIG. 1B also shows an override 27 that can be used to force a "safe" condition when one of the sensors is in an unsafe state. This circuit is optional but, when provided, allows the driver to drive to a mechanic with a failing sensor. A counter 28 or other abuse preventer prevents the override from being used more than a predetermined number times before repairing the sensor (for example three times). The counter 28 can be optionally reset whenever the sensor logic produces a safe signal. In the present example, the driver can enter a special PIN code 29 to activate the override and override the sensors. While providing this circuit lowers the

5

overall safety threshold of the system slightly, the act of performing the override can be made difficult enough that a driver will not routinely use it to avoid having to fix a faulty sensor. Optionally, the override could be restricted to use only by a certified mechanic. In this case, the driver would not be allowed to operate it. While a simple override switch can be used, for additional safety, a PIN or barcode or any other unique identification 29 might be required. If a PIN is used, a PIN entry method can be used such as a keypad or a card swipe. If a barcode is used, a small barcode reader can be supplied.

FIG. 2 shows a male tank adapter 2 on the vehicle with a coupled (usually spring coupled) female filling locking coupler 3 and high pressure feed hose 1. The feed hose 1 typically originates at a compressor or storage tank. A proximity sensor 4 with electrical connections 5 senses the presence of the feed hose (unsafe condition). The female locking coupler 3 typically has a connect/disconnect fitting that clamps the tank adapter 2 and makes a high pressure, leak-proof connection. It should be noted that the female filling locking coupler 3 could be a quick-release, break-away fitting. This would add a final mechanical precaution to the system where the hose would manually break free in the case that somehow the rest of the system failed.

FIG. 3 shows the same setup but with the feed hose 1 and the female locking coupler 3 uncoupled and withdrawn (safe position) from the male filling adapter 2.

FIG. 4 shows a fill cavity 6 and fill access lid 7 that swings open on hinges. A second proximity sensor 8 has a second electrical feed 9. This sensor 8 can be used in embodiments of the invention either alone or in conjunction with the feed hose sensor 4. An optional microswitch 20 is also shown.

Any type of sensor can be used with the present invention. A preferred sensor is a magnetic or optical proximity sensor; however, other sensor types can be used such as ultra-sonic sensors and others.

FIG. 5 shows a different arrangement of the fueling system. This type of system is more commonly found on trucks and larger vehicles. A panel 13 opens and closes to allow access to a tank adapter 12. An isolation valve 10 can be manually operated or can operate in conjunction with a lever 16 on the panel 13. A third proximity sensor 14 with electrical feed 15 can be used to sense whether the panel is open or closed (and hence, if the lever 16 is used with the valve 10, whether the valve is open or closed). The valve 10 isolates the tank adapter 12 from the tank when the panel is closed.

FIG. 6 shows the same embodiment as FIG. 5 except that an optional microswitch 23 has been added to the panel 13. This is a simple mechanical backup that does not rely on proximity sensors. It should be noted that any of the proximity sensors shown in the various embodiments of the present invention may be replaced by microswitches or other mechanical devices, or each proximity sensor may be optionally backed up with an additional microswitch. In addition, FIG. 6 shows an optional additional microswitch 21 on the isolation valve 10.

It should be noted that an optional natural gas leak sensor can also be included with the system of the present invention to provide an addition source of safety. Such a sensor could keep the vehicle in a disabled state if an unreasonable concentration of free natural gas is sensed. Such a sensor could also sound an alarm on the vehicle or on the compressor. It is also possible that any of the above-mentioned sensors or circuits can communicate with the lockout or other logic wirelessly such as by radio or light. Finally, it should be noted that a processor with memory and a stored

6

program can perform the logic function of the lockout, override and/or abuse preventer. This can also be any digital or analog logic circuit or simply relays. In addition, an internet or network interface can be provided to remotely report or log the status of the system. This feature can be useful for fleet operators to track safety, for example, the number of attempts to drive off under a lockout condition or the number of overrides.

The present invention provides a multiply-redundant system to enhance the safety of home or business fueling of compressed natural gas vehicles. It can also be used on vehicles using propane or any other compressed gas fuel. A system of one or more sensors determines if the fueling system is in a safe state. This can be a state with the high pressure fill hose removed, the fueling compartment access lid closed, and any isolation valve in the correct position. A lockout prevents either the vehicle from starting or otherwise makes it impossible to move the vehicle when the system is not in a safe state. In order that a driver can get a faulty sensor repaired, an override can be provided that allows the safety sensors to be overridden and the vehicle moved. This override can be equipped with an abuse preventer that only allows the override to be used for a predetermined number of times before the override itself is disabled. This prevents a driver from putting off getting a faulty sensor repaired. An audio and/or visual indicator can sound or display when the driver inserts the key into the ignition in an unsafe state, or alternatively, when the driver turns the key to start.

In the present invention, a lockout is any method, means or technique to prevent a vehicle from moving including a circuit or module that can disable the ignition or transmission or disable the vehicle in any other way. An override is any method, means or technique to allow the vehicle to move in spite of the lockout—any way of overriding the lockout. An abuse preventer is any method, means or technique to prevent abuse of the override by limiting the number of times (especially successive times) it can be used.

Several descriptions and illustrations have been presented to aid in understanding the features of the present invention. One with skill in the art will realize that numerous changes and variations may be made without departing from the spirit of the invention. Each of these changes and variations is within the scope of the present invention.

The invention claimed is:

1. A vehicle safety system for disabling a vehicle while fueling the vehicle, wherein the vehicle is powered by a compressed or liquefied fuel, comprising:

a pressurized fuel supply tank carrying the compressed or liquefied fuel and located external to the vehicle, and a pressurized fuel receiving tank located within the vehicle;

a break-away fitting attached to a fuel fill hose coupled to the pressurized fuel supply tank during fueling of the vehicle, the break-away fitting configured to allow separation of the fuel fill hose from the pressurized fuel supply tank in case of an unexpected vehicle pull away;

a proximity sensor adapted to sense a presence of the fuel fill hose in proximity to the pressurized fuel receiving tank, the proximity sensor in communication with a logic circuit;

the logic circuit adapted to produce a safe electrical signal indicating a safe condition when the fuel fill hose is not in proximity to the pressurized fuel receiving tank;

a lockout circuit adapted to mechanically or electrically disable the vehicle when the safe electrical signal is absent; and

7

a manual override controlled by human intervention and overriding the logic circuit to produce the safe electrical signal when activated.

2. The vehicle safety system of claim 1, wherein the safe signal is communicated by wireless communication to the
5
lockout circuit.

3. The vehicle safety system of claim 2, wherein the wireless communication is transmitted by radio or light.

4. The vehicle safety system of claim 1, wherein the manual override includes a switch, PIN code entry, barcode
10
entry or card swipe.

5. The vehicle safety system of claim 1, wherein the override is adapted to be used only a predetermined number of successive times.

6. The vehicle safety system of claim 4, further comprising a counter circuit adapted to count successive uses of the
15
override, and wherein the counter circuit is reset when the logic circuit produces the safe electrical signal.

7. The vehicle safety system of claim 1, wherein the proximity sensor is magnetic, optical or ultra-sonic.

8. The vehicle safety system of claim 1, wherein the fuel
20
comprises compressed or liquefied propane.

9. The vehicle safety system of claim 1, wherein the fuel comprises compressed or liquefied natural gas.

10. The vehicle safety system of claim 1, further comprising a network enabling remote reporting of a status of the
25
system.

11. A vehicle safety system for disabling a vehicle during fueling, wherein the vehicle is powered by compressed or
30
liquefied fuel supplied to the vehicle through a fuel fill hose, and the vehicle has an ignition and transmission, the vehicle safety system comprising:

a pressurized fuel supply tank carrying the compressed or liquefied fuel and located external to the vehicle, and a pressurized fuel receiving tank located within the vehicle;

8

a break-away fitting attached to the fuel fill hose, the fuel fill hose in communication with the pressurized fuel receiving tank and being coupled to the pressurized fuel supply tank during fueling of the vehicle the one or more break-away fittings configured to allow separation of the fuel fill hose from the pressurized fuel supply tank in case of an unexpected vehicle pull-away; a proximity sensor adapted to sense a presence of the fuel fill hose in proximity to the pressurized fuel receiving tank, the proximity sensor in communication with a logic circuit;

the logic circuit adapted to produce a safe electrical signal indicating a safe condition when the fuel fill hose is not in proximity to the pressurized fuel receiving tank;

a lockout circuit adapted to disable the vehicle when the safe electrical signal is absent; and

a manual override controlled by human intervention and overriding the logic circuit to produce the safe electrical signal when activated.

12. The vehicle safety system of claim 11, wherein the compressed or liquefied fuel comprises propane.

13. The vehicle safety system of claim 11, wherein the compressed or liquefied fuel comprises natural gas.

14. The vehicle safety system of claim 1, wherein the pressurized fuel receiving tank includes a fill cap, and further comprising a microswitch adapted to sense the position of the fill cap, and wherein the logic circuit fails to produce the safe electrical signal when the microswitch senses the fill cap is open.

15. The vehicle safety system of claim 1, further comprising a tank fill adaptor, an isolation valve for isolating the tank fill adaptor from the pressurized fuel receiving tank, and a sensor for sensing whether the isolation valve is open or closed.

* * * * *